

Lecture 14

Operators on Hilbert Space

$\text{spec}(T)$ is the spectrum; $T^\dagger$ the adjoint; inner products follow the physics convention. A ★ marks a harder problem.

Bounded operators, adjoints, spectrum

Exercise 14.1 (computational) Find the operator norm and the adjoint of multiplication by $m(x) = e^{ix}$ on $L^2[0, 2\pi]$.

Exercise 14.2 (computational) The diagonal operator $De_n = \frac{n}{n+1}e_n$ on ℓ^2 . Find its point spectrum and full spectrum.

Exercise 14.3 (computational) Let $De_n = 2^{-n}e_n$ on ℓ^2 , where $e_1, e_2, \dots$ is the standard orthonormal basis ($n \geq 1$). Find $\|D\|$, the adjoint $D^\dagger$, the point spectrum, and the full spectrum; is D self-adjoint? compact?

Exercise 14.4 (computational) Find the operator norm, the adjoint, and the spectrum of multiplication by $g(x) = 2x$ on $L^2[0, 1]$. Does it have eigenvalues?

Exercise 14.5 (computational) Let S be the right shift $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$ on ℓ^2 . Compute $S^\dagger S$ and $SS^\dagger$, give $\|S\|$, and decide whether S is an isometry and whether it is unitary.

Exercise 14.6 (computational) For $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ on $\mathbb{C}^2$, find the adjoint $A^\dagger$, classify A (self-adjoint / unitary / normal), and give $\text{spec}(A)$.

Exercise 14.7 (★) Show that multiplication by x on $L^2[0, 1]$ has no eigenvalues, yet $\text{spec}(\hat{x}) = [0, 1]$.

Exercise 14.8 (proof) Show that a bounded diagonal operator $De_n = d_n e_n$ on ℓ^2 (with $\sup_n |d_n| < \infty$) has $\|D\| = \sup_n |d_n|$.

Position, momentum, and the commutator

Exercise 14.9 (proof) Verify $[\hat{x}, \hat{p}]\psi = i\hbar\psi$ for $\hat{p} = -i\hbar d/dx$ on a smooth ψ .

Exercise 14.10 (proof) Prove no finite matrices X, P satisfy $[X, P] = i\hbar I$.

Exercise 14.11 (computational) Using $[A, BC] = [A, B]C + B[A, C]$, compute $[\hat{x}, \hat{p}^2]$.

Exercise 14.12 (computational) Using $[A, BC] = [A, B]C + B[A, C]$ and $[\hat{x}, \hat{p}] = i\hbar$, compute $[\hat{p}, \hat{x}^2]$.

Exercise 14.13 (computational) Compute $[\hat{x}, \hat{p}^3]$.

Exercise 14.14 (computational) Compute $[\hat{x}^2, \hat{p}^2]$, leaving the answer in terms of $\hat{x}$ and $\hat{p}$.

Exercise 14.15 (computational) In natural units ($\hbar = 1$, so $[\hat{x}, \hat{p}] = i$) set $a = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p})$ and $a^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p})$. Compute $[a, a^\dagger]$.

Exercise 14.16 (computational) With $a, a^\dagger$ as above and the number operator $N = a^\dagger a$, use $[a, a^\dagger] = \text{id}$ to compute $[N, a]$.

Exercise 14.17 (conceptual) (*The boundary condition is part of the observable.*) Let $T = -\frac{d^2}{dx^2}$ act on twice-differentiable functions on $[0, 1]$ with the L^2 inner product.

1. Integrating by parts twice, show that

$$\langle Tu, v \rangle - \langle u, Tv \rangle = \left[\overline{u} v' - \overline{u'} v \right]_0^1.$$

2. Conclude that T is symmetric ($\langle Tu, v \rangle = \langle u, Tv \rangle$) on the Dirichlet domain $u(0) = u(1) = 0$, and explain why the same operator with *no* boundary condition is not.

Exercise 14.18 (proof) Prove $[\hat{x}, \hat{p}^n] = n i \hbar \hat{p}^{n-1}$ for all $n \geq 1$ by induction.

The spectral theorem and dynamics

Exercise 14.19 (computational) For the compact self-adjoint T with $Te_n = \frac{1}{n}e_n$ on ℓ^2 , write Tv and $\|Tv\|^2$ for $v = \sum_n c_n e_n$.

Exercise 14.20 (computational) For $H = \frac{\hbar\omega}{2}\sigma_z$, evolve $|\psi\rangle(0) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ to time t .

Exercise 14.21 (computational) Measuring σ_x on $|0\rangle$: give the outcome probabilities and $\langle\sigma_x\rangle$.

Exercise 14.22 (computational) Let $Te_n = \frac{1}{n}e_n$ on ℓ^2 (compact, self-adjoint) and $v = 3e_1 + 4e_2$. Compute Tv , $\|v\|^2$, $\|Tv\|^2$, $\langle v, Tv \rangle$, and the Rayleigh quotient $\langle v, Tv \rangle / \|v\|^2$.

Exercise 14.23 (computational) A three-level system has $H|n\rangle = n\varepsilon|n\rangle$ for $n = 0, 1, 2$. Evolve $|\psi\rangle(0) = \frac{1}{\sqrt{3}}(|0\rangle + |1\rangle + |2\rangle)$ to time t .

Exercise 14.24 (computational) Measuring σ_z in $|\psi\rangle = \frac{1}{2}(\sqrt{3}|0\rangle + |1\rangle)$: give the outcome probabilities and $\langle\sigma_z\rangle$.

Exercise 14.25 (computational) Using the spectral decomposition $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, compute $e^{i\alpha\sigma_z}$ for real α .

Exercise 14.26 (★) For a density operator ρ , prove $\langle A \rangle = \text{tr}(\rho A)$ reduces to $\langle\psi, A\psi\rangle$ when $\rho = |\psi\rangle\langle\psi|$ is pure.

Exercise 14.27 (proof) For a self-adjoint $H = \sum_n E_n |e_n\rangle\langle e_n|$ with real E_n and orthonormal $\{e_n\}$, show that $U(t) = e^{-iHt/\hbar}$ acts by $|e_n\rangle \mapsto e^{-iE_n t/\hbar} |e_n\rangle$ and is unitary.